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Through mold interconnect drill feature

Abstract

Embodiments disclosed herein include electronic packages. In an embodiment, an electronic package comprises a package substrate, a first die electrically coupled to the package substrate, and a mold layer over the package substrate and around the first die. In an embodiment, the electronic package further comprises a through mold opening through the mold layer, and a through mold interconnect (TMI) in the through mold opening, wherein a center of the TMI is offset from a center of the through mold opening.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8222538	12/2011	Yoshida	174/262	H05K 3/4007
2011/0140268	12/2010	Cheah	N/A	N/A
2012/0211885	12/2011	Choi	257/737	H01L 25/105
2012/0280404	12/2011	Kwon	N/A	N/A
2014/0138826	12/2013	Kumar	N/A	N/A
2014/0353821	12/2013	Yu	257/737	H01L 25/105
2019/0006319	12/2018	Karhade	N/A	H01L 23/3135

OTHER PUBLICATIONS

Office Action from U.S. Appl. No. 18/374,585, mailed Jun. 12, 2024, 13 pgs. cited by applicant
Office Action from U.S. Appl. No. 18/374,585, mailed Oct. 8, 2024, 12 pgs. cited by applicant
Office Action from U.S. Appl. No. 18/374,585, mailed Feb. 5, 2025, 16 pgs. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/550,773, filed on Aug. 26, 2019, the entire contents of which is hereby incorporated by reference herein.

TECHNICAL FIELD
(1) Embodiments of the present disclosure relate to semiconductor devices, and more particularly to vents for through mold interconnects (TMIs).
BACKGROUND
(2) Advanced packaging architectures may include multi-chip packages. In some embodiments, package-on-package (PoP) architectures are used to couple a first package (e.g., a system-on-a-chip (SoC) package) to a second package (e.g., a memory package). The two packages are coupled together by through mold interconnects (TMIs). The TMIs include solder balls that are exposed by drilling a via through the mold layer. Typically, the via to ball spacing is constrained with the solder ball contacting the via walls. This is done to retain as much mold material as possible to provide mechanical integrity to the package, minimize warpage concerns, and prevent solder balls from shorting during the attach process. During attachment of the second package (and in subsequent reflow operations such as surface mount processes), outgassing of moisture absorbed by the

package organic materials (particularly outgassing by a capillary underfill (CUF) material) in the first package exerts a high vapor pressure on the solder balls due to the constrained gap between the via sidewalls and the solder. This can potentially result in the solder being separated from the bottom pad. Such a phenomenon may be referred to in the art as “ball lifting”. Ball lifting causes an electrical open and results in yield loss.
(3) One solution to reduce the forces that cause ball lifting may include increasing the TMI drill diameter to provide a larger space for outgassing moisture. However, in architectures with tight spacing between the TMI and the die (e.g., approximately 500 μm or less), increases in the TMI drill diameter are not possible. Closer to the die, the CUF fillet extends close to the TMI, and a larger drill diameter would require ablating both the mold material and the CUF, which is not practical. Furthermore, at the package edges, a larger drill diameter risks the loss of mechanical integrity of the mold.
(4) Other embodiments may include providing outgassing paths down through the package substrate of the first package. This is suboptimal because it requires forming openings through the various metal layers of the package substrate. Accordingly, electrical performance is sacrificed in order to reduce ball lifting during assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a cross-sectional illustration of an electronic system that includes a first package and a second package in a package-on-package (PoP) architecture with a vent for the through mold interconnect (TMI), in accordance with an embodiment.
(2) FIG. **2**A is a plan view illustration of the through mold opening and a vent, in accordance with an embodiment.
(3) FIG. **2**B is a plan view illustration of the through mold opening and a vent with a smaller diameter than the through mold opening, in accordance with an embodiment.
(4) FIG. **2**C is a cross-sectional illustration of the through mold opening and a pair of vents, in accordance with an embodiment.
(5) FIG. **3**A is a cross-sectional illustration of a package with a solder ball embedded in a mold layer, in accordance with an embodiment.
(6) FIG. **3**B is a cross-sectional illustration of the package after an opening is formed through the mold layer, in accordance with an embodiment.
(7) FIG. **3**C is a cross-sectional illustration of the package after a vent is formed through the mold layer, in accordance with an embodiment.
(8) FIG. **3**D is a plan view illustration of the package in FIG. **3**C, in accordance with an embodiment.
(9) FIG. **4**A is a cross-sectional illustration of an electronic package with a plurality of rows of TMIs with the TMI in the first row including a vent, in accordance with an embodiment.
(10) FIG. **4**B is a plan view illustration of the electronic package in FIG. **4**A, in accordance with an embodiment.
(11) FIG. **4**C is a cross-sectional illustration of an electronic package with a plurality of rows of TMIs with vents being formed along more than one row, in accordance with an embodiment.
(12) FIG. **5** is a cross-sectional illustration of an electronic system that comprises a PoP architecture with TMIs that include vents, in accordance with an embodiment.
(13) FIG. **6** is a schematic of a computing device built in accordance with an embodiment.
EMBODIMENTS OF THE PRESENT DISCLOSURE
(14) Described herein are electronic packages with vents for through mold interconnects (TMIs), in accordance with various embodiments. In the following description, various aspects of the

illustrative implementations will be described using terms commonly employed by those skilled in the art to convey the substance of their work to others skilled in the art. However, it will be apparent to those skilled in the art that the present invention may be practiced with only some of the described aspects. For purposes of explanation, specific numbers, materials and configurations are set forth in order to provide a thorough understanding of the illustrative implementations. However, it will be apparent to one skilled in the art that the present invention may be practiced without the specific details. In other instances, well-known features are omitted or simplified in order not to obscure the illustrative implementations.

(15) Various operations will be described as multiple discrete operations, in turn, in a manner that is most helpful in understanding the present invention, however, the order of description should not be construed to imply that these operations are necessarily order dependent. In particular, these operations need not be performed in the order of presentation.

(16) As noted above, compact electronic systems that include the use of package-on-package (PoP) architectures often suffer from ball lifting defects during assembly. It has been shown that moisture outgassing from organic materials in a package assembly (particularly from the capillary underfill (CUF) material) is the primary driving force behind the ball lifting. Accordingly, embodiments disclosed herein provide a vent alongside the through mold interconnects (TMIs) that allows for the outgassed moisture to escape through the mold layer and eliminates the scenario of a high vapor pressure building up against the solder balls.

(17) Such vent architectures are particularly useful because they are compatible with the tight spacing of advanced packaging architectures. For example, for TMIs adjacent to the CUF with a tight keep-out zone, the vent may be formed on an edge of the TMI that is opposite from the CUF. Accordingly, there is no problem with needing to ablate both mold material and CUF material. That is, existing laser processes may be used to form the vents. Furthermore, the vents also provide an outgassing path that does not need to pass through the package substrate. Accordingly, there is no need to redesign the conductive layers of the package substrate to accommodate outgassing pathways. Therefore, the conductive layers can be optimized for electrical performance without the need to consider outgassing issues.

(18) Additionally, it has been found that the outgassing that causes ball lifting primarily affects the interior TMIs (i.e., the TMIs closest to the die and CUF since the CUF is the primary source of outgassed moisture). Accordingly, embodiments may include providing the vent feature only on the interior TMIs. That is, the TMIs in rows outside the innermost row can be formed with standard processing without the need for vents. However, it is to be appreciated that embodiments may also include vents formed in other rows (i.e., proximate to the package edge or on internal rows of TMI) without significantly compromising the mechanical integrity of the package and losing the positive benefits of mold in package warpage control.

(19) Referring now to FIG. **1**, a cross-sectional illustration of an electronic package **100** is shown, in accordance with an embodiment. In an embodiment, the electronic package **100** may comprise a PoP architecture. That is, the electronic package **100** may comprise a first package **130** and a second package **120**. In an embodiment, the first package **130** may also be referred to as a bottom package, a system-on-a-chip (SoC) package, a multi-chip package (MCP), or the like. In an embodiment, the second package **120** may be referred to as a top package, a memory package, or the like.

(20) In an embodiment the second package **120** may comprise one or more dies **122**. For example, two dies **122** are shown in FIG. **1**. In the case where a plurality of dies **122** are used, the dies **122** may be stacked over each other. The stacked dies **122** may be offset from each other to allow for wire bonds **124** to provide electrical connection to a pad **125** on a side of a solder resist layer **121** opposite from the dies **122**. In an embodiment, a mold layer **123** may embed the one or more dies **122**. In a particular embodiment, the one or more dies **122** are memory dies. However, it is to be appreciated that the dies **122** may comprise any type of die.

(21) In an embodiment, the first package **130** may comprise a package substrate **131**. The package substrate **131** may include layers of insulating material (e.g., build-up layers, etc.) with conductive features (e.g., traces, vias, pads, etc.). For simplicity, the plurality of layers and the majority of the conductive features are omitted. However, a pad **133** below the solder ball **143** is shown. The pad **133** may be electrically coupled to the one or more dies **134**/**135** by conductive traces in the package substrate **131**. In an embodiment, the first die **134** may be electrically coupled to the package substrate **131** by interconnects **136**. In an embodiment, the interconnects **136** are shown as a single copper bump. However, it is to be appreciated that the interconnects **136** may comprise any suitable interconnect architecture. In some embodiments, the interconnects **136** may be referred to generally as first level interconnects (FLIs). In an embodiment, the second die **135** may be electrically coupled to the first die **134** by interconnects **136**.
(22) In an embodiment, the first die **134** comprises transistors that are fabricated at a first process node, and the second die **135** comprises transistors that are fabricated at a second process node that is more advanced than the first process node. However, it is to be appreciated that the first die **134** and the second die **135** may also be formed at the same process node in some embodiments. Furthermore, while two dies **134**/**135** are shown, it is to be appreciated that in some embodiments the first package **130** may include single die. Other embodiments may include a plurality of first dies **134** and/or a plurality of second dies **135**.
(23) In an embodiment, a capillary underfill (CUF) material **137** may be disposed under the first die **134**. The CUF material **137** may surround the interconnects **136** below the first die **134**. The CUF material **137** may also extend out away from the edge of the first die **134**. For example, the CUF material **137** may extend towards the solder ball **143**. In an embodiment, the second die **135** may be embedded in a mold layer **138**. Other embodiments may include embedding (or partially embedding) the second die **135** with the CUF material **137**.
(24) In an embodiment, a mold layer **132** is disposed over a surface of the package substrate **131**. The mold layer **132** may embedded and/or surround the first dies **134** and the second die **135**. In and embodiment, the mold layer **132** directly contacts portions of the CUF material **137**. The mold layer **132** is a different material than the CUF material **137**. In a particular embodiment, the CUF material **137** is a material that more readily absorbs moisture compared to the mold layer **132**. Accordingly, during attachment of the second package **120**, significant outgassing from the CUF material **137** occurs.
(25) In an embodiment, the first package **130** is electrically coupled to the second package **120** by one or more through mold interconnects (TMIs) **147**. The TMIs **147** may comprise the solder ball **143** and a solder ball **144**. While shown as two distinct balls, it is to be appreciated that the solder ball **143** may merge with the solder ball **144** during attachment of the second package **120** to the first package **130**. Accordingly, an electrical path from pad **125** of the second package **120** to the pad **133** of the first package **130** is provided by the TMI **147**.
(26) In an embodiment, the TMI **147** passes through a through mold opening **140**. In an embodiment, the through mold opening **140** comprises a first opening **141** and a second opening **142**. The first opening **141** may extend down into the mold layer **132** and land on a surface of the solder ball **143**. In an embodiment, the first opening **141** may be augmented by the addition of a second opening **142**. The second opening **142** may also be referred to herein as a vent. The second opening **142** passes entirely through the mold layer **132**. Accordingly, the second opening **142** may expose a portion of the surface of the package substrate **131**.
(27) In an embodiment, the second opening **142** is on an edge of the TMI **147** that is opposite from the die **134**. Accordingly, the additional second opening **142** is spaced away from the CUF material **137**. Accordingly, the formation of the second opening **142** only requires drilling through the mold layer **132**. Due to the tight spacing D between the die **134** and the TMI **147**, forming a vent between the TMI **147** and the die **134** is not a feasible solution. Particularly, since the laser is optimized for ablating the mold layer **132**, it is difficult to form a vent that passes through both the

CUF material **137** and the mold layer **132**. In an embodiment, the spacing D may be approximately 500 μm or less.

(28) The second opening **142** provides a path around the TMI **147** through which moisture can vent. For example, a path **139** of moisture from the CUF material **137** through mold layer **132** and out the second opening **142** is shown, in accordance with an embodiment. While the path **139** shown in FIG. **1** passes through the TMI **147**, it is to be appreciated that the path **139** would typically wrap around the TMI **147** (out of the plane of FIG. **1**) instead of directly passing through the TMI **147** as shown.

(29) In the cross-sectional illustration shown in FIG. **1**, the through mold opening **140** appears as a single continuous opening. For example, the first opening **141** defines a first edge of the opening and the second opening **142** defines a second edge of the opening. From such a perspective, the center of the TMI **147** will be offset from a center of the through mold opening **140**. Particularly, the center of the TMI **147** will be shifted towards the die **134**. However, it is to be appreciated that in some embodiments (and from different perspectives), the distinction between the first opening **141** and the second opening **142** is more distinguishable.

(30) Referring now to FIGS. **2**A-**2**C, a series of plan view illustrations of a portion of the mold layer **232** is shown, in accordance with various embodiments. The view shown in FIG. **2**A-**2**C more clearly illustrates the intersection of the first opening **241** and the second opening **242**. In FIGS. **2**A-**2**C the underlying layers (e.g., a pad and the package substrate) and the TMI are omitted in order to not obscure the illustration of the through mold opening **240**.

(31) Referring now to FIG. **2**A, a plan view illustration of a portion of the mold layer **232** is shown, in accordance with an embodiment. In an embodiment, the mold layer **232** includes a through mold opening **240**. As shown, the through mold opening **240** includes a first opening **241** and a second opening **242**. The first opening **241** may have a first diameter D.sub.1 and second opening **242** may have a second diameter D.sub.2. In the illustrated embodiment, the first diameter D.sub.1 and the second diameter D.sub.2 may be substantially equal to each other. However, as will be described above, the first diameter D.sub.1 may be different than the second diameter D.sub.2.

(32) As shown, the second opening **242** intersects the first opening **241**. For example, an offset O between the center point of the second opening **242** relative to the center point of the first opening **241** may be approximately 30 μm or less. In a particular embodiment, the offset O may be approximately 10 μm or less. Depending on the amount of the offset O, the shape of the through mold opening **240** will change. For example, the through mold opening **240** may appear oval shaped at low offsets O, and an indent may appear as the offset O increases.

(33) Referring now to FIG. **2**B, a plan view illustration of a portion of a mold layer **232** is shown, in accordance with an additional embodiment. As shown, the through mold opening **240** includes a first opening **241** and a second opening **242** that intersects the first opening **241**. In an embodiment, the first opening **241** comprises a first diameter D.sub.1 and the second opening **242** comprises a second diameter D.sub.2. In an embodiment, the second diameter D.sub.2 is different than the first diameter D.sub.1. Particularly, the second diameter D.sub.2 is shown as being smaller than the first diameter D.sub.1.

(34) Referring now to FIG. **2**C, a plan view illustration of a portion of a mold layer **232** is shown, in accordance with an additional embodiment. In an embodiment, a through mold opening **240** may be formed through the mold layer **232**. The through mold opening **240** may comprise a first opening **241** and a plurality of second openings **242**. For example two second openings **242**.sub.A and **242**.sub.B are shown. In an embodiment, the second openings **242** may each intersect the first opening **241**. The first opening **241** may have a first diameter D.sub.1 and the second openings **242** may have a second diameter D.sub.2 that is different than the first diameter D.sub.1. For example, the second diameter D.sub.2 is shown as being smaller than the first diameter D.sub.1. Furthermore, while both second openings **242** are shown with substantially the same diameter D.sub.2, it is to be appreciated that the second openings **242** need not have the same diameter.

Additionally, the second openings **242**.sub.A and **242**.sub.B may have different offsets from the first opening **241**.

(35) Referring now to FIGS. **3**A-**3**D, a series of illustrations depicting a process for forming a through mold opening **340** in an electronic package **330** is shown, in accordance with an embodiment. In FIGS. **3**A-**3**D, only a portion of the electronic package **330** is shown for simplicity. However, it is to be appreciated that the electronic package **330** may be similar to the first package **130** in FIG. **1**.

(36) Referring now to FIG. **3**A, a cross-sectional illustration of a portion of an electronic package **330** is shown, in accordance with an embodiment. In the illustrated portion, a package substrate **331** and a mold layer **332** over the package substrate **331** is shown. The mold layer **332** may embed a solder ball **343**. In an embodiment, the solder ball **343** is positioned over a pad **333**. The pad **333** may be electrically coupled to one or more dies (not shown) of the electronic package **330** through conductive features in the package substrate **331**.

(37) Referring now to FIG. **3**B, a cross-sectional illustration of the electronic package **330** after a first opening **341** is formed into the mold layer **332** is shown, in accordance with an embodiment. In an embodiment, the first opening **341** may expose a portion of the solder ball **343**. In a particular embodiment, the first opening **341** intersects the solder ball **343**. That is, the sidewalls **348** of the first opening **341** terminate on a surface of the solder ball **343**. Accordingly, the first opening **341** does not extend all the way through a thickness of the mold layer **332**. In an embodiment, the centerline of the solder ball **343** may be substantially aligned with a centerline of the first opening **341**.

(38) In an embodiment, the first opening **341** may be formed with any suitable process. For example, the first opening **341** may be formed with a laser drilling process. As is typical with laser drilling, the sidewalls **348** may have a tapered profile. That is, a diameter at the top of the first opening **341** may be greater than a diameter at the bottom of the first opening **341**.

(39) Referring now to FIG. **3**C, a cross-sectional illustration of the electronic package **330** after a second opening **342** is formed into the mold layer **332** is shown, in accordance with an embodiment. In an embodiment, the second opening **342** may extend all the way through a thickness of the mold layer **332**. That is, the sidewall **349** of the second opening **342** may terminate at the surface of the package substrate **331**. In an embodiment, the second opening **342** exposes a portion of the surface of the package substrate **331**.

(40) In an embodiment, the second opening **342** is offset from the first opening **341** and intersects the first opening **341** to form a through mold opening **340**. That is, the through mold opening **340** comprises the first opening **341** and the second opening **342**. In an embodiment, the addition of the second opening **342** results in a centerline of the solder ball **343** being offset from a centerline of the through mold opening **340**. Additionally, the second opening **342** provides a venting path that allows for moisture or the like to vent out of the electronic package **330** without providing enough pressure on the solder ball **343** to cause ball lifting.

(41) Referring now to FIG. **3**D, a plan view illustration of the electronic package **330** in FIG. **3**C is shown, in accordance with an embodiment. The sidewalls **348** and **349** of the first opening **341** and the second opening **342** are visible in the plan view due to their tapered profile. The exposed surface of the package substrate **331** is also clearly visible. In a particular embodiment, the exposed surface of the package substrate **331** may be crescent shaped. However, the shape of the exposed surface of the package substrate may vary, depending on the shapes, sizes, offsets, etc. of the first opening **341** and the second opening **342**. In an embodiment, a portion of the pad **333** may also be exposed. In other embodiments, the solder ball **343** may entirely cover the pad **333** and it will not be visible.

(42) Referring now to FIGS. **4**A and **4**B, a cross-sectional illustration and a corresponding plan view illustration of an electronic package **430** is shown, in accordance with an additional embodiment. The electronic package **430** may be similar to the first electronic package **130** in FIG.

1, with the exception that a plurality of solder balls **443**.sub.A-C and a plurality of through mold openings **440**.sub.A-C are included. For example, the electronic package **430** comprises a package substrate **431**, a first die **434**, a second die **435**, a CUF material **437** and mold layers **432** and **438**.
(43) In an embodiment, the solder balls **443**.sub.A-C may each be positioned on a pad **433**. The first row of solder balls **443**.sub.A is considered the innermost row of solder balls **443**. That is, the first solder balls **443**.sub.A are closer to the die **434** than the second row of solder balls **443**B and the third row of solders ball **443**.sub.C. Furthermore, while three rows of solder balls **443** are shown, it is to be appreciated that any number of rows of solder balls **443** may be included depending on the needs of the electronic package **430**. In an embodiment, each of the solder balls **443** may be substantially similar to each other. That is, the volume of solder in each solder ball **443**.sub.A in the first row may be substantially similar to the volume of solder in each solder ball **443** in any of the other rows.
(44) In an embodiment, the through mold openings **440** are not uniform for each of the rows. Particularly, the through mold openings **440**.sub.A in the first row are larger than the through mold openings **440**.sub.B-C in the second row and the third row. For example, the through mold openings **440**.sub.A may comprise a first opening **441** and a second opening **442** that intersects the first opening **441**. In an embodiment, centerlines of the first solder balls **443**.sub.A may be offset from the centerline of the through mold opening **440**.sub.A. This allows for the second opening **442** to expose a portion of the package substrate **431**.
(45) In an embodiment, the second opening **442** is located along an edge of the solder ball **443**.sub.A that is opposite from the die **434**. This allows for existing laser ablation processes to be used to form the second opening **442**. Particularly, the second opening **442** only passes through the mold layer **432**, so there is no need for adjusting the laser ablation process from the one used to form the first opening **441**. If the second opening were to be formed on the side of the solder ball **443**.sub.A closest to the die **434**, then the laser ablation process would need to account for the presence of the CUF material **437**, and require additional refinement of the laser ablation process.
(46) Accordingly, outgassing from the CUF material **437** may escape up through the second opening **442** of the first through mold openings **440**.sub.A. Since the pressure from outgassed moisture is relieved at the first through mold openings **440**.sub.A, there is not sufficient pressure in the subsequent rows to produce ball lifting. Accordingly, in some embodiments, only the first row of through mold openings **440**.sub.A include a second opening (vent) **442**.
(47) However, it is to be appreciated that second openings **442** may optionally be added to any of the rows of through mold openings **440**. FIG. **4**C provides an example of such an embodiment. As shown, a second openings **442** may also be formed along the outermost row of solder balls **443**.sub.C. For example, the through mold openings **440**.sub.C may comprise a first opening **441** and a second opening **442** that intersects the first opening **441**. In an embodiment, the second opening **442** may be positioned towards the interior of the package **430** in order to maintain structural integrity.
(48) Referring now to FIG. **5**, a cross-sectional illustration of an electronic system **500** is shown, in accordance with an embodiment. In an embodiment, the electronic system **500** comprises a board **570**, a first electronic package **530**, and second electronic package **520**.
(49) In an embodiment, the board **570** may be any suitable board. For example, the board **570** may be a mother board, a printed circuit board (PCB) or the like. In an embodiment, the board **570** is electrically coupled to the first electronic package **530**. For example, the board **570** may be electrically coupled to the first electronic package **530** with any suitable interconnect architecture **571**. For example the interconnect architecture **571** is shown as being solder balls, but other interconnect architectures (e.g., sockets, wire bonds, etc.) may be used.
(50) In an embodiment, the first electronic package **530** may comprise a package substrate **531**. The package substrate **531** is electrically coupled to the board **570**. In an embodiment, a plurality of dies **534**/**535** may be electrically coupled to the package substrate **531**. In an embodiment, a first

die **534** may be a base die onto which a plurality of second dies **535** are attached. The first die **534** may be formed at a first process node, and the second dies **535** may be formed at a second process node that is more advanced than the first process node. In an embodiment, a CUF material **537** may surround the first die **534**. In an embodiment, the CUF material **537** may extend out laterally towards TMIs **540**. In an embodiment, a mold layer **532** may be disposed over the package substrate **531** and around the dies **534**/**535**.

(51) In an embodiment, the plurality of dies **534**/**535** may be electrically coupled to pads **533** onto which TMIs **540** are positioned. In an embodiment, the TMIs may comprise a solder ball **543** that has merged with a second solder ball **544**. In an embodiment, the innermost TMIs **540** (i.e., the TMIs **540** closest to the die **534**) may further comprise a vent **542**. The vent **542** is part of a through mold opening that provides access to the solder ball **543**. In an embodiment, the vent **542** may extend all the way through the mold layer **532** so that a portion of the package substrate **531** is exposed. In an embodiment, the vent **542** provides a pathway through which outgassed moisture from the CUF material **537** escapes during the attachment of the second electronic package **520**. Accordingly, ball lifting is eliminated since the vent reduces pressure on the bottom surface of the solder balls **543**. In an embodiment, the vent **542** is separated from the die **534** by the solder ball **543**. Positioning the vent **542** opposite from the die **534** prevents the vent **542** from needing to be drilled through the CUF material **537**. Accordingly, the spacing between the innermost TMIs **540** and the die **534** can be minimized. For example, the spacing between the innermost TMIs **540** and the die **534** may be approximately 500 μm or less. In some embodiments, only the innermost TMIs **540** include a vent **542**.

(52) In an embodiment, the second electronic package **520** is electrically coupled to the first electronic package by the TMIs **540**. In an embodiment, the second electronic package **520** may comprise one or more dies **522** that are electrically coupled to the first electronic package **530** by wire bonds **524** or any other suitable interconnect. A solder resist layer **521**, redistribution layers or the like may be below the one or more dies **522**. In an embodiment, a mold layer **523** may embed the wire bonds **524** and the one or more dies **522**.

(53) FIG. **6** illustrates a computing device **600** in accordance with one implementation of the invention. The computing device **600** houses a board **602**. The board **602** may include a number of components, including but not limited to a processor **604** and at least one communication chip **606**. The processor **604** is physically and electrically coupled to the board **602**. In some implementations the at least one communication chip **606** is also physically and electrically coupled to the board **602**. In further implementations, the communication chip **606** is part of the processor **604**.

(54) These other components include, but are not limited to, volatile memory (e.g., DRAM), non-volatile memory (e.g., ROM), flash memory, a graphics processor, a digital signal processor, a crypto processor, a chipset, an antenna, a display, a touchscreen display, a touchscreen controller, a battery, an audio codec, a video codec, a power amplifier, a global positioning system (GPS) device, a compass, an accelerometer, a gyroscope, a speaker, a camera, and a mass storage device (such as hard disk drive, compact disk (CD), digital versatile disk (DVD), and so forth).

(55) The communication chip **606** enables wireless communications for the transfer of data to and from the computing device **600**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a non-solid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not. The communication chip **606** may implement any of a number of wireless standards or protocols, including but not limited to Wi-Fi (IEEE 802.11 family), WiMAX (IEEE 802.16 family), IEEE 802.20, long term evolution (LTE), Ev-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPRS, CDMA, TDMA, DECT, Bluetooth, derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The computing device **600** may include a plurality of communication chips **606**. For instance, a first

communication chip **606** may be dedicated to shorter range wireless communications such as Wi-Fi and Bluetooth and a second communication chip **606** may be dedicated to longer range wireless communications such as GPS, EDGE, GPRS, CDMA, WiMAX, LTE, Ev-DO, and others.

(56) The processor **604** of the computing device **600** includes an integrated circuit die packaged within the processor **604**. In some implementations of the invention, the integrated circuit die of the processor **604** may be part of an electronic package that comprises a TMI with a vent opposite from the processor **604**, in accordance with embodiments described herein. The term “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory.

(57) The communication chip **606** also includes an integrated circuit die packaged within the communication chip **606**. In accordance with another implementation of the invention, the integrated circuit die of the communication chip **606** may be part of an electronic package that comprises a TMI with a vent opposite from the communication chip **606**, in accordance with embodiments described herein.

(58) The above description of illustrated implementations of the invention, including what is described in the Abstract, is not intended to be exhaustive or to limit the invention to the precise forms disclosed. While specific implementations of, and examples for, the invention are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize.

(59) These modifications may be made to the invention in light of the above detailed description. The terms used in the following claims should not be construed to limit the invention to the specific implementations disclosed in the specification and the claims. Rather, the scope of the invention is to be determined entirely by the following claims, which are to be construed in accordance with established doctrines of claim interpretation.

(60) Example 1: an electronic package, comprising: a package substrate; a first die electrically coupled to the package substrate; a mold layer over the package substrate and around the first die; a through mold opening through the mold layer; and a through mold interconnect (TMI) in the through mold opening, wherein a center of the TMI is offset from a center of the through mold opening.

(61) Example 2: the electronic package of Example 1, wherein the through mold opening comprises: a first opening; and a second opening, wherein the second opening intersects the first opening.

(62) Example 3: the electronic package of Example 2, wherein a diameter of the first opening is substantially equal to a diameter of the second opening.

(63) Example 4: the electronic package of Example 2, wherein a diameter of the first opening is different than a diameter of the second opening.

(64) Example 5: the electronic package of Examples 1-4, wherein the first die is separated from the second opening by the TMI.

(65) Example 6: the electronic package of Examples 1-5, wherein an edge of the TMI nearest the first die is spaced away from the first die by approximately 500 μm or less.

(66) Example 7: the electronic package of Examples 1-6, wherein the through hole opening exposes a surface of the package substrate.

(67) Example 8: the electronic package of Examples 1-7, further comprising: a second package electrically coupled to the first die by the TMI.

(68) Example 9: the electronic package of Examples 1-8, further comprising: a plurality of second dies over the first die.

(69) Example 10: the electronic package of Examples 1-9, further comprising: a capillary underfill (CUF) material under the first die and between the TMI and the first die.

(70) Example 11: an electronic package comprising: a package substrate; a die on the package

substrate; a mold layer over the package substrate; a row of first through mold interconnects (TMIs) adjacent to an edge of the die, wherein each first TMI comprises a vent; and a row of second TMIs separated from the edge of the die by the row of first TMIs.
(71) Example 12: the electronic package of Example 11, wherein the vents are positioned along an edge of the first TMIs opposite from the edge of the die.
(72) Example 13: the electronic package of Example 11 or Example 12, wherein the vents expose a surface of the package substrate.
(73) Example 14: the electronic package of Examples 11-13, wherein the TMIs comprise: a solder interconnect; and an opening into the mold layer.
(74) Example 15: the electronic package of Example 14, wherein the vents intersect the opening.
(75) Example 16: the electronic package of Examples 11-15, further comprising: a capillary underfill (CUF) layer under the die and between the edge of the die and the row of first TMIs.
(76) Example 17: the electronic package of Examples 11-16, wherein the row of first TMIs is spaced away from the edge of the die by approximately 500 μm or less.
(77) Example 18: the electronic package of Examples 11-17, further comprising: a second package coupled to the row of first TMIs and the row of second TMIs.
(78) Example 19: an electronic system, comprising: a first package, wherein the first package comprises: a package substrate; a mold layer over the package substrate; a through mold interconnect (TMI), wherein the TMI comprises a vent; and a second package electrically coupled to the first package by the TMI.
(79) Example 20: the electronic system of Example 19, wherein the first package comprises a plurality of dies.
(80) Example 21: the electronic system of Example 20, wherein the second package comprises a memory die.
(81) Example 22: the electronic system of Examples 19-21, wherein the vent exposes a surface of the package substrate.
(82) Example 23: the electronic system of Examples 19-22, wherein the TMI further comprises: a first opening, wherein the first opening has a first diameter, and wherein the vent intersects the first opening, and wherein the vent comprises an arc having the first diameter.
(83) Example 24: the electronic system of Examples 19-23, further comprising: a second TMI, wherein the second TMI does not include a vent.
(84) Example 25: the electronic system of Examples 19-24, further comprising: a board electrically coupled to the package substrate.

Claims

1. An electronic package, comprising: a first die coupled to a package substrate, the first die having a top surface; a second die coupled to a first portion of the top surface of the first die, the second die having a top surface; a first mold layer over a second portion of the top surface of the first die, the first mold layer laterally adjacent to and in direct contact with an entirety of a sidewall of the second die, and the first mold layer having a top surface co-planar with the top surface of the second die; a second mold layer on the package substrate, the second mold layer laterally adjacent to and in contact with the first mold layer, and the second mold layer having a top surface co-planar with the top surface of the first mold layer; a through mold opening in the second mold layer, the through mold opening having a center and a top; and a through mold interconnect in the through mold opening, the through mold interconnect having a portion above the top of the through mold opening, the portion of the through mold interconnect having a center offset from the center of the through mold opening, wherein the through mold interconnect is laterally spaced apart from both the first die and the second die.
2. The electronic package of claim 1, wherein the center of the through mold interconnect is offset

from the center of the through mold opening in a direction toward the first die and the second die.
3. The electronic package of claim 1, wherein the opening has a first sloped sidewall and a second sloped sidewall from a cross-sectional perspective.
4. The electronic package of claim 3, wherein the through mold interconnect is in contact with the first sloped sidewall but is not in contact with the second sloped sidewall.
5. The electronic package of claim 1, further comprising: a second through mold interconnect in a second through mold opening, wherein the through mold interconnect is laterally between the second through mold interconnect and the first die.
6. Electronic package of claim 5, wherein the second through mold interconnect has a center substantially aligned with a center of the second through mold opening.
7. The electronic package of claim 1, wherein the through mold interconnect is on a contact pad exposed by the through mold opening.
8. The electronic package of claim 7, wherein the contact pad has a center offset from the center of the through mold opening.
9. An electronic package, comprising: a first die coupled to a package substrate, the first die having a top surface; a second die coupled to a first portion of the top surface of the first die, the second die having a top surface; a first mold layer over a second portion of the top surface of the first die, the first mold layer laterally adjacent to and in direct contact with an entirety of a sidewall of the second die, and the first mold layer having a top surface co-planar with the top surface of the second die; a second mold layer on the package substrate, the second mold layer laterally adjacent to and in contact with the first mold layer, and the second mold layer having a top surface co-planar with the top surface of the first mold layer; a through mold opening in the second mold layer, the through mold opening having a center and a top; a contact pad exposed by the through mold opening, the contact pad having a center offset from the center of the through mold opening; and a through mold interconnect in the through mold opening and on the contact pad, the through mold interconnect having a portion above the top of the through mold opening, the portion of the through mold interconnect having a center offset from the center of the through mold opening, wherein the through mold interconnect is laterally spaced apart from both the first die and the second die.
10. The electronic package of claim 9, wherein the center of the contact pad is offset from the center of the through mold opening in a direction toward the first die and the second die.
11. The electronic package of claim 9, wherein the opening has a first sloped sidewall and a second sloped sidewall from a cross-sectional perspective.
12. The electronic package of claim 11, wherein the through mold interconnect is in contact with the first sloped sidewall but is not in contact with the second sloped sidewall.
13. The electronic package of claim 9, further comprising: a second through mold interconnect in a second through mold opening, wherein the through mold interconnect is laterally between the second through mold interconnect and the first die.
14. The electronic package of claim 13, wherein the second through mold interconnect has a center substantially aligned with a center of the second through mold opening.
15. A system comprising: a first electronic package, comprising: a first die coupled to a package substrate, the first die having a top surface; a second die coupled to a first portion of the top surface of the first die, the second die having a top surface; a first mold layer over a second portion of the top surface of the first die, the first mold layer laterally adjacent to and in contact with an entirety of a sidewall of the second die, and the first mold layer having a top surface co-planar with the top surface of the second die; a second mold layer on the package substrate, the second mold layer laterally adjacent to and in direct contact with the first mold layer, and the second mold layer having a top surface co-planar with the top surface of the first mold layer; a through mold opening in the second mold layer, the through mold opening having a center and a top; and a through mold interconnect in the through mold opening, the through mold interconnect having a portion above the top of the through mold opening, the portion of the through mold interconnect having a center

offset from the center of the through mold opening, wherein the through mold interconnect is laterally spaced apart from both the first die and the second die; and a second electronic package coupled to the first electronic package by the through mold interconnect, wherein the second electronic package comprises a third die, the third die vertically over the second die of the first electronic package.

16. The system of claim 15, wherein the center of the through mold interconnect is offset from the center of the through mold opening in a direction toward the first die and the second die.

17. The system of claim 15, wherein the opening has a first sloped sidewall and a second sloped sidewall from a cross-sectional perspective.

18. The system of claim 17, wherein the through mold interconnect is in contact with the first sloped sidewall but is not in contact with the second sloped sidewall.

19. The system of claim 15, further comprising: a second through mold interconnect in a second through mold opening, wherein the through mold interconnect is laterally between the second through mold interconnect and the first die.

20. The system of claim 19, wherein the second through mold interconnect has a center substantially aligned with a center of the second through mold opening.

21. The system of claim 15, wherein the through mold interconnect is on a contact pad exposed by the through mold opening.

22. The system of claim 21, wherein the contact pad has a center offset from the center of the through mold opening.